

CHAPTER 10

KINETIC MODEL OF

MATTER

2.1 Kinetic particle model of matter

2.1.1 States of matter

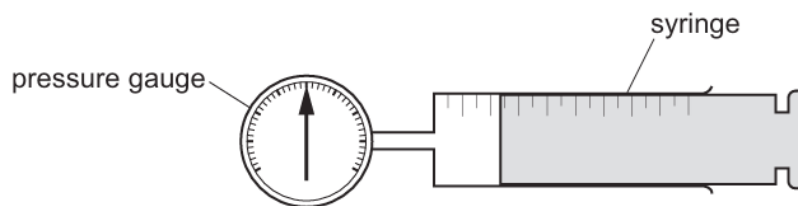
- 1 Know the distinguishing properties of solids, liquids and gases
- 2 Know the terms for the changes in state between solids, liquids and gases (gas to solid and solid to gas transfers are **not** required)

2.1.2 Particle model

- 1 Describe, qualitatively, the particle structure of solids, liquids and gases, relating their properties to the forces and distances between particles and to the motion of the particles (atoms, molecules, ions and electrons)
- 2 Describe the relationship between the motion of particles and temperature, including the idea that there is a lowest possible temperature ($-273\text{ }^{\circ}\text{C}$), known as absolute zero, where the particles have least kinetic energy
- 3 Describe the pressure and the changes in pressure of a gas in terms of the forces exerted by particles colliding with surfaces, creating a force per unit area
- 4 Explain qualitatively, in terms of particles, the relationship between:
 - (a) pressure and temperature at constant volume
 - (b) volume and temperature at constant pressure
 - (c) pressure and volume at constant temperature
- 5 Recall and use the equation $p_1V_1 = p_2V_2$, including a graphical representation of the relationship between pressure and volume for a gas at constant temperature

1. O/N/23/11/Q14

Some gas is trapped in a large syringe.



The atmospheric pressure is 100 kPa. The gas pressure is 200 kPa above atmospheric pressure.

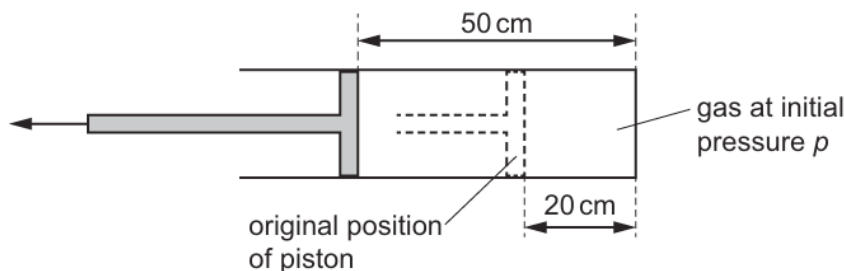
The piston moves outwards and the volume of the trapped gas doubles. The temperature remains constant.

What is the new gas pressure?

- A** 100 kPa **B** 150 kPa **C** 200 kPa **D** 400 kPa

2. O/N/23/12/Q19

A gas is trapped inside a cylinder by a movable piston. The length of the gas column is 20 cm and the pressure inside the cylinder is p .



The piston is pulled out a distance of 30 cm, so that the length of the gas column is now 50 cm. The temperature of the gas does not change.

What is the new pressure of the gas?

- A** $0.40p$ **B** $0.60p$ **C** $1.5p$ **D** $2.5p$

3. M/J/23/11/Q14

Which row describes the shape and the volume of a liquid at constant temperature?

	shape	volume
A	fixed	fixed
B	fixed	not fixed
C	not fixed	fixed
D	not fixed	not fixed

4. M/J/23/11/Q15

Which row shows how the forces and the distances between the particles in a solid generally compare with the forces and distances in a liquid?

	forces between particles in a solid	distances between particles in a solid
A	stronger	greater
B	stronger	smaller
C	weaker	greater
D	weaker	smaller

5. M/J/23/12/Q14

What is a property of both liquids and gases?

- A** They always fill their containers.
- B** They are incompressible.
- C** They can flow.
- D** They have particles in fixed positions.

6. M/J/23/12/Q15

In an experiment, the volume of a gas in a cylinder is measured as the pressure of the gas is increased.



The values obtained are shown.

pressure P/Pa	1.0×10^5	2.0×10^5	3.0×10^5	4.0×10^5
volume V/m^3	20×10^{-5}	10×10^{-5}	6.0×10^{-5}	4.0×10^{-5}

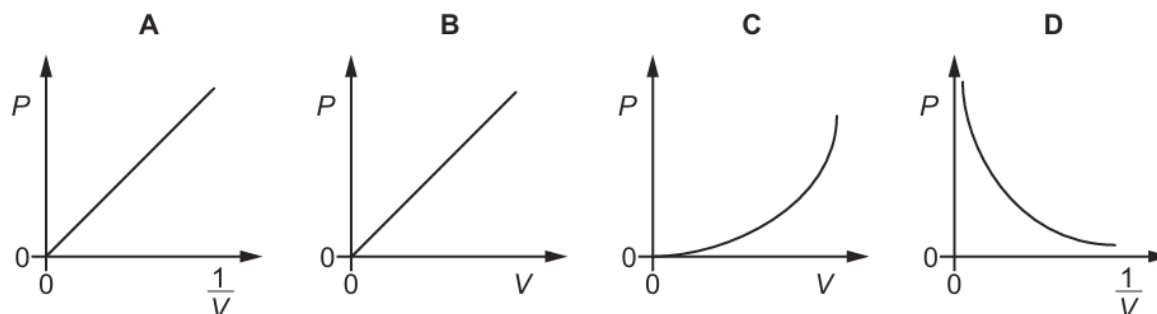
What is a possible explanation for these results?

- A** After the pressure is doubled, gas starts leaking into the cylinder.
- B** After the pressure is doubled, gas starts leaking out of the cylinder.
- C** The temperature of the gas is constant.
- D** The temperature of the gas is increasing.

7. O/N/22/11/Q14

The pressure P of some trapped air is varied. The mass and the temperature of the trapped air remain constant.

Which graph shows how the volume V of the air varies with the pressure P ?



8. O/N/22/12/Q15

A gas syringe contains a volume V_1 of gas at pressure p_1 .

The volume is reduced to V_2 without any change of temperature.

What is the expression for the **change** in pressure?

A $\frac{p_1 V_1}{V_2} + p_1$
 B $p_1 + \frac{p_1 V_2}{V_1}$
 C $\frac{p_1 V_1}{V_2} - p_1$
 D $\frac{p_1 V_2}{V_1} - p_1$

9. M/J/22/11/Q18

Which row describes the arrangement of particles and the forces between particles in a solid?

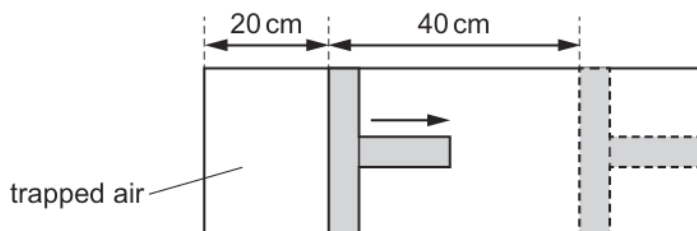
	arrangement of particles	forces between particles
A	regular	strong
B	regular	weak
C	irregular	strong
D	irregular	weak

10. O/N/21/11/Q9

Air is trapped in a cylinder by a piston. The pressure of the air is p and the length of the air column is 20 cm.

The piston is moved outwards. The length of the air column increases by 40 cm.

The temperature of the air remains constant.



What is the new air pressure?

A $\frac{p}{2}$

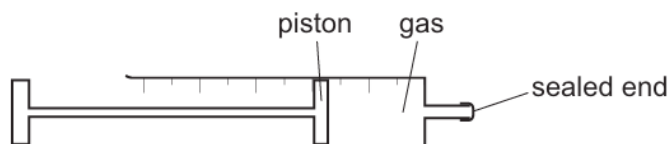
B $\frac{p}{3}$

C $2p$

D $3p$

11. M/J/21/11/Q10

A sealed gas syringe contains a fixed mass of gas.



The piston is moved and the volume of the gas doubles. The temperature of the gas does not change.

What happens to the pressure of the gas?

A halves

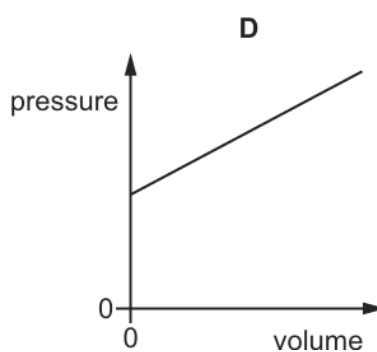
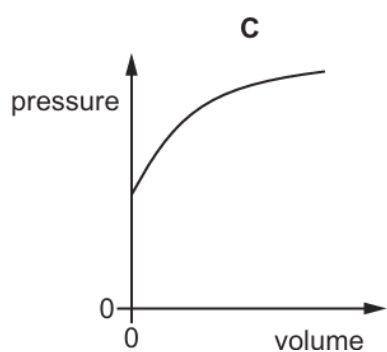
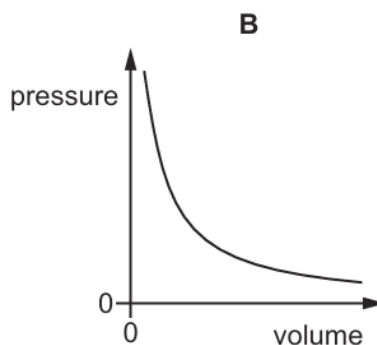
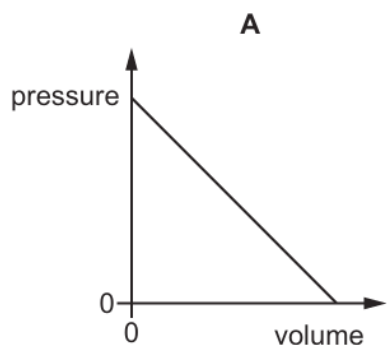
B no change

C doubles

D triples

12. M/J/21/12/Q14

Which graph shows the relationship between the pressure and the volume of a fixed mass of gas at constant temperature?



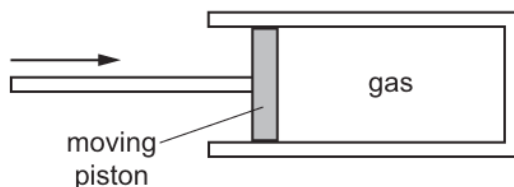
13. O/N/20/12/Q21

Which description of a liquid is correct?

- A** fixed shape, fixed volume
- B** fixed shape, no fixed volume
- C** no fixed shape, fixed volume
- D** no fixed shape, no fixed volume

14. O/N/21/12/Q22

A mass of gas is trapped inside a cylinder by a moving piston.



As the piston moves in, the volume of the gas decreases but the temperature stays the same.

What happens to the gas pressure, and why?

	pressure	reason
A	decreases	the molecules have less room to move in the cylinder
B	decreases	the molecules move more slowly between collisions
C	increases	the molecules collide more frequently with the piston
D	unchanged	the temperature of the gas is constant

15. M/J/20/11/Q24

Which row explains why a liquid has a fixed volume but does **not** have a fixed shape?

	force between molecules in the liquid	movement of molecules in the liquid
A	large	free to move within the liquid
B	large	vibrate at fixed positions
C	small	free to move within the liquid
D	small	vibrate at fixed positions

16. M/J/20/11/Q25

A fixed mass of gas is enclosed in a cylinder by a piston which is free to move.

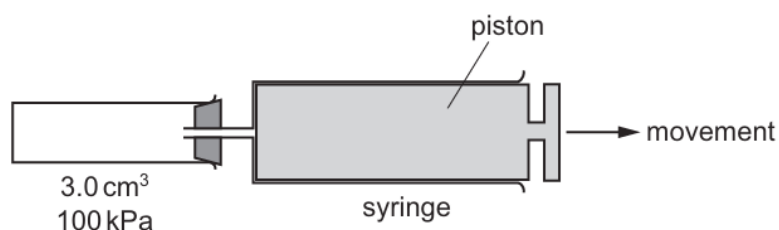
Which combination of changes to the pressure and to the temperature **must** increase the density of the gas?

	pressure	temperature
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

17. M/J/20/12/Q15

A small vessel of volume 3.0 cm^3 contains air at a pressure of 100 kPa .

The small vessel is connected to a syringe. The piston is fully inserted into the syringe.



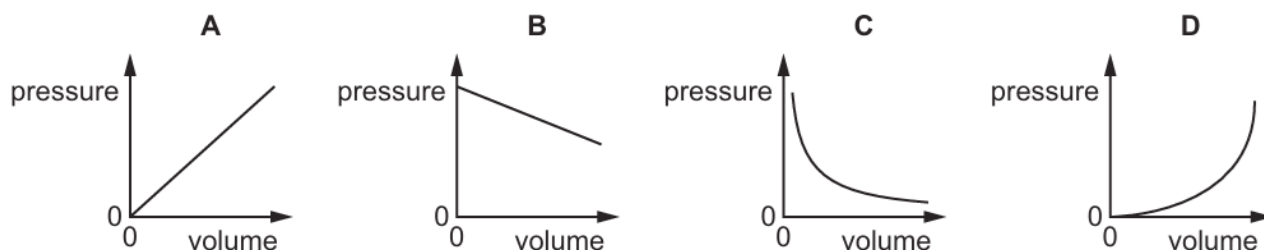
The piston is moved slowly to the right so that the air in the syringe has a volume of 12 cm^3 . The temperature of the air does not change.

What is the pressure of the air in the syringe?

- A** 20 kPa **B** 25 kPa **C** 80 kPa **D** 100 kPa

18. O/N/19/11/Q16

Which graph shows the relationship between the pressure and the volume of a fixed mass of gas at constant temperature?



19. M/J/19/11/Q21

Which statement about particles in a liquid is correct?

- A** They all have the same speed.
B They are stationary.
C They move at random.
D They vibrate about a fixed point.

20. M/J/19/12/O15

The pressure on the gas contained in a sealed syringe is slowly doubled. The volume of the gas halves.

What **must** be true about the density and the temperature of the gas?

	density	temperature
A	decreases	increases
B	increases	decreases
C	increases	remains constant
D	remains constant	increases

21. O/N/18/11/Q12

Some gas is trapped in a closed container. The gas is cooled and the volume of the container is kept constant.

What happens to the gas molecules?

- A** They collide with the walls more often.
- B** They contract.
- C** They get closer together.
- D** They move more slowly.

22. M/J/18/12/Q19

A sealed container of gas is heated and the pressure inside increases.

What happens to the molecules of the gas to cause this increase in pressure?

- A** Their kinetic energy decreases.
- B** They become heavier.
- C** They expand.
- D** They hit the container more frequently.

23. O/N/17/11/Q11

A gas is heated in a closed container of constant volume.

What happens to the molecules of the gas?

- A** They collide with the walls with less force.
- B** They expand.
- C** They move faster.
- D** They move further apart.

24. O/N/17/12/Q11

A gas syringe contains a fixed mass of air. The volume of the air is 240 cm^3 and it exerts a pressure of $5.0 \times 10^4\text{ Pa}$. The air is slowly compressed, keeping the temperature constant, until the pressure is $1.5 \times 10^5\text{ Pa}$.

What is the final volume of air?

- A** 72 cm^3 **B** 80 cm^3 **C** 720 cm^3 **D** 800 cm^3

25. O/N/17/12/Q12

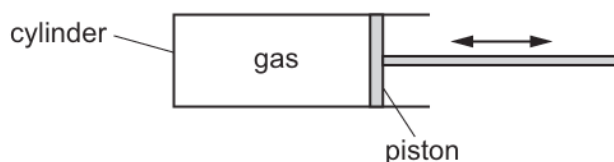
An extremely large increase in pressure is needed to compress a liquid. A gas can be compressed by a much smaller increase in pressure.

Which statement explains this?

- A** Molecules repel each other very strongly when very close.
B The attractive forces between molecules are small at large distances.
C The molecules in the gas collide with the walls of their container and rebound.
D The molecules of a liquid are constantly moving at random.

26. M/J/17/12/Q20

A fixed mass of gas is enclosed in a cylinder with a movable piston.



The gas is initially at pressure p_1 and has a volume V_1 .

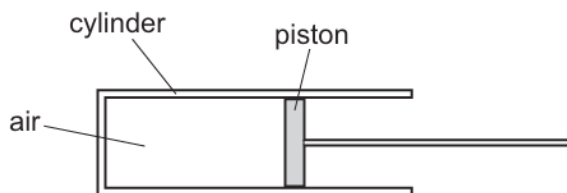
The temperature is kept constant. The piston is moved so that the pressure becomes p_2 and the volume becomes V_2 .

Which equation is correct?

- A** $\frac{p_1}{V_1} = \frac{p_2}{V_2}$
B $\frac{p_1}{p_2} = \frac{V_1}{V_2}$
C $p_1 V_1 = p_2 V_2$
D $p_1 V_2 = p_2 V_1$

27. O/N/16/11/Q14

Air is trapped in a cylinder by a piston.



The piston is pushed further into the cylinder.

How do the pressure and the volume of the air in the syringe change?

	pressure	volume
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

28. O/N/16/11/Q15

Which statement explains how a pressure is exerted by a gas on a container?

- A** Gas molecules collide with other gas molecules in the container.
- B** Gas molecules collide with the walls of the container.
- C** Gas molecules exert strong attractive forces on each other.
- D** Gas molecules exert strong repulsive forces on each other.

29. O/N/16/12/Q18

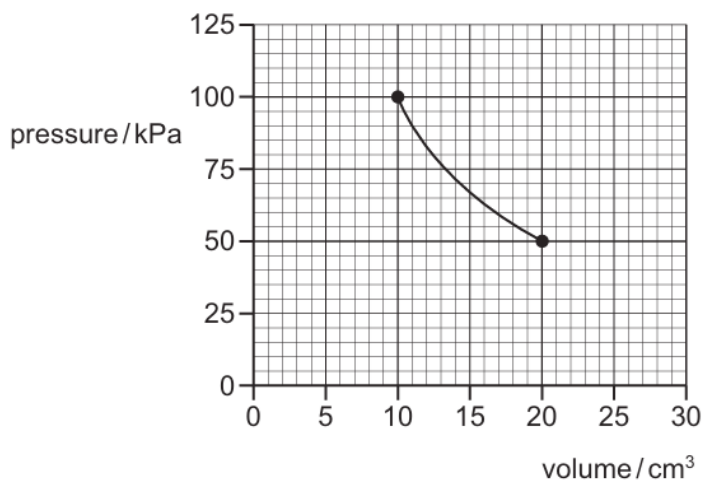
A gas expands slowly and its temperature remains constant.

What happens to the gas molecules?

- A** They move further apart and their average speed decreases.
- B** They move further apart and their average speed increases.
- C** They move further apart and their average speed remains unchanged.
- D** They stay the same distance apart and their average speed decreases.

30. M/J/15/12/Q11

The graph shows how the pressure of a fixed mass of gas varies with volume at constant temperature.



What is the volume of the gas when the pressure is 25 kPa?

- A** 2.5 cm³ **B** 10 cm³ **C** 30 cm³ **D** 40 cm³

31. M/J/15/12/Q20

Some gas is trapped in a closed container. The gas is cooled and the volume of the container is kept constant.

What happens to the gas molecules?

- A** They collide with the walls more often.
B They contract.
C They get closer together.
D They move more slowly.

32. M/J/14/11/Q14

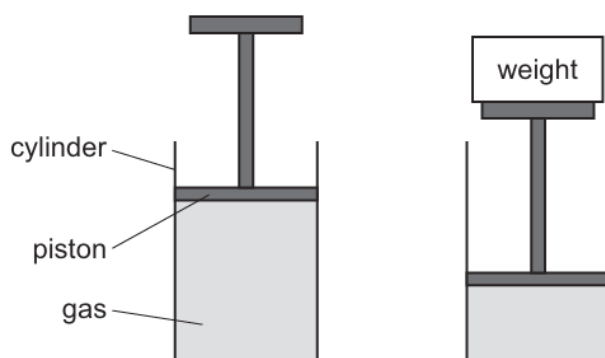
A gas occupies a volume of 2.0 m³ in a cylinder at a pressure of 240 kPa. A piston compresses the gas until the volume is 0.50 m³, the temperature remaining constant.

What is the new pressure of the gas?

- A** 60 kPa **B** 240 kPa **C** 480 kPa **D** 960 kPa

33. M/J/14/11/Q18

A piston is supported by gas trapped in a cylinder.



A weight is put on the piston. The volume of gas supporting the piston decreases but the temperature of the gas is unchanged.

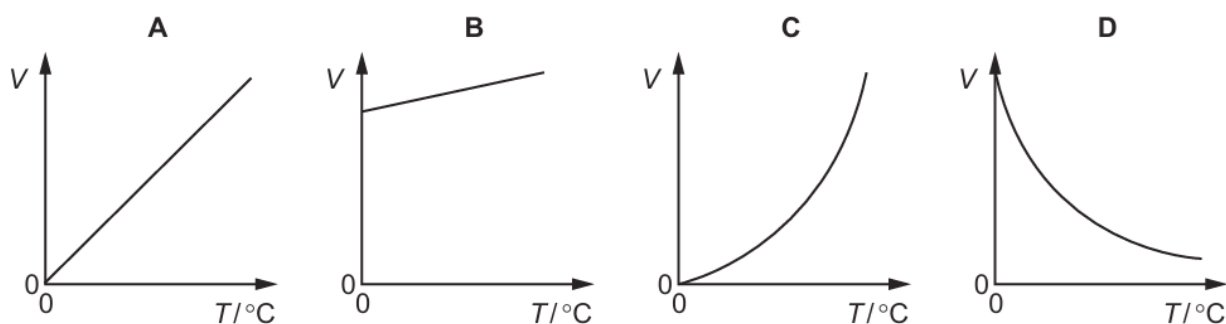
What happens to the molecules?

- A** They hit the piston more frequently.
- B** They move more slowly.
- C** They have a smaller size.
- D** They have more kinetic energy.

34. M/J/14/11/Q21

A fixed mass of gas in a syringe at 0°C is heated at constant pressure.

Which graph shows the variation of volume V with temperature T , measured in $^{\circ}\text{C}$?



35. O/N/13/11/Q18

Which row correctly describes the shape and volume of a gas?

	shape	volume
A	fixed	fixed
B	fixed	not fixed
C	not fixed	fixed
D	not fixed	not fixed

36. O/N/13/12/Q12

The conditions at the bottom and at the surface of a lake are given in the table.

	bottom of lake	surface of lake
temperature	10 °C	10 °C
pressure	500 kPa	100 kPa

A bubble of volume 1.0 cm³ forms at the bottom of the lake.

What is the volume of the bubble as it reaches the surface?

- A** 0.20 cm³ **B** 0.25 cm³ **C** 4.0 cm³ **D** 5.0 cm³

37. O/N/13/12/Q19

A fixed mass of gas is kept at constant pressure. Its temperature is raised.

What happens to the volume of the gas and to its molecules?

	volume	molecules
A	decreases	move more slowly
B	increases	stay the same distance apart
C	increases	move further apart
D	no change	move at the same speed

38. O/N/13/12/Q20

Air is heated in a sealed container with constant volume.

Why does the air pressure increase when the temperature increases?

- A** The air molecules expand.
B The air molecules bounce off each other more frequently.
C The air molecules bounce off the walls more frequently.
D The number of air molecules increases.

39. M/J/13/11/Q16

When a gas in a container of fixed volume is heated, the pressure of the gas increases.

Which statement explains this?

- A** The molecules expand and get heavier, so they hit each other harder.
B The molecules have less room to move, so they collide with each other more frequently.
C The molecules hit the walls harder and less frequently.
D The molecules move faster, so they collide with the walls more frequently.

40. M/J/13/12/Q12

An airtight container holds a fixed mass of gas. Its pressure and volume are measured on four occasions when the temperature is 20 °C.

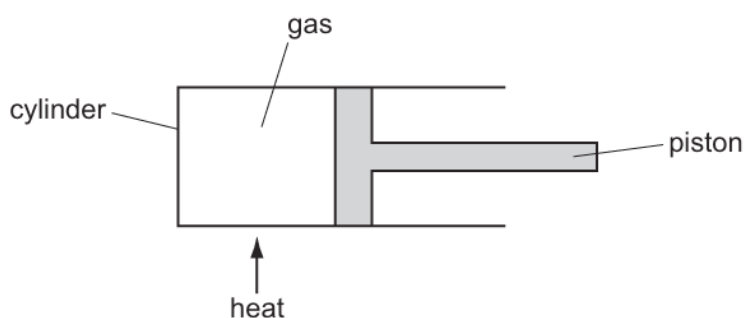
The results are shown in the table. Three sets of readings are correct.

Which set of readings is **not** correct?

	pressure / kPa	volume / cm ³
A	120	36
B	100	48
C	80	60
D	60	80

41. M/J/13/12/Q20

The diagram shows a fixed mass of gas in a cylinder fitted with a piston that can move easily.



What is the change, if any, in the pressure and volume of the gas after it is heated?

	pressure	volume
A	no change	increases
B	decreases	no change
C	decreases	increases
D	increases	decreases

42. O/N/12/11/Q12

Four syringes contain air at atmospheric pressure.

The air in all four syringes is slowly compressed. The results are given in the table.

Which syringe shows the smallest pressure change?

	volume before compression / cm ³	volume after compression / cm ³
A	50	10
B	100	50
C	400	25
D	400	100

43. O/N/12/11/Q15

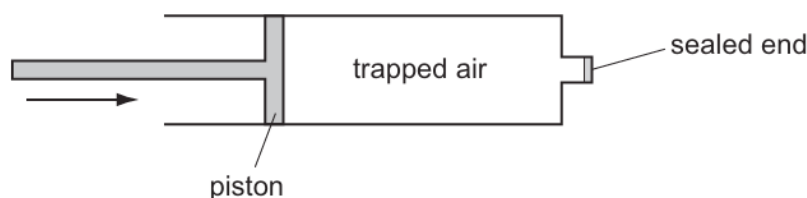
When a gas is rapidly compressed to a smaller volume, its temperature increases.

What happens to the gas molecules?

- A** They move closer together and their average speed decreases.
- B** They move closer together and their average speed increases.
- C** They move closer together and their average speed remains unchanged.
- D** They stay the same distance apart and their average speed increases.

44. O/N/12/12/Q12

The end of a bicycle pump is sealed.



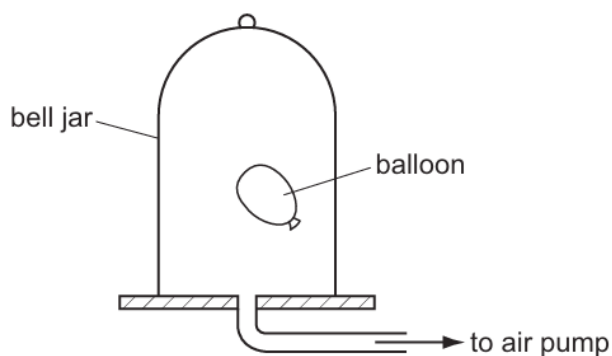
The temperature of the air does not change as the piston is pushed in. The final volume of trapped air is $\frac{1}{4}$ of the original volume.

How does the pressure of the trapped air change?

- A** It decreases to $\frac{1}{4}$ of the original value.
- B** It decreases to $\frac{3}{4}$ of the original value.
- C** It increases to 3 times the original value.
- D** It increases to 4 times the original value.

45. M/J/12/11/Q12

A partially-inflated balloon is placed inside a bell jar. The bell jar is connected to an air pump.



The air pump is switched on and air is removed from the bell jar.

What happens to the pressure and to the volume of the gas inside the balloon?

	pressure	volume
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

46. M/J/12/11/Q15

A gas is enclosed in a container of fixed volume. It gains heat energy from an external source.

What happens to the molecules of the gas?

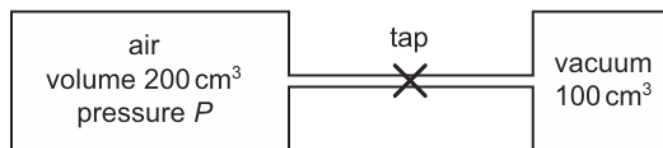
- A** They expand.
- B** They move faster inside the container.
- C** They move further apart.
- D** They vibrate with greater frequency.

47. M/J/12/12/Q12

Two cylinders are connected by a thin pipe.

One cylinder has a volume of 200 cm^3 and contains air at pressure P .

The other cylinder has a volume of 100 cm^3 and contains a vacuum (no air).



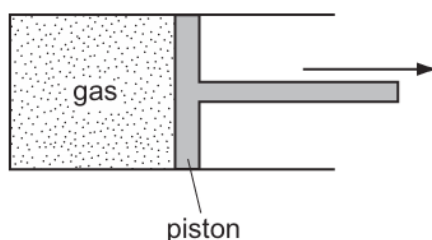
Initially, the tap is closed.

What is the final pressure of the air after the tap is opened?

- A** $\frac{P}{2}$ **B** $\frac{2P}{3}$ **C** $\frac{3P}{2}$ **D** $2P$

48. M/J/12/12/Q15

A fixed mass of gas is enclosed in a cylinder by a movable piston.



The piston is moved so that the volume occupied by the gas increases. The temperature remains constant.

What happens to the pressure of the gas and why does this happen?

	pressure	reason
A	decreases	the molecules move more slowly
B	decreases	the molecules collide with the piston less frequently
C	increases	the molecules move more quickly
D	increases	the molecules collide with the piston more frequently

49. M/J/11/11/Q15

According to the kinetic theory, matter is made up of very small particles in a constant state of motion.

Which row best describes the particle behaviour in the liquid state?

	forces between particles	motion of particles
A	strong	move randomly at high speed
B	strong	vibrate but are free to move position
C	strong	vibrate to and fro around a fixed position
D	weak	move randomly at high speed

50. O/N/09/Q17

A quantity of gas is trapped in a container by a piston exerting a force F .

The temperature of the gas is raised while F remains unchanged.

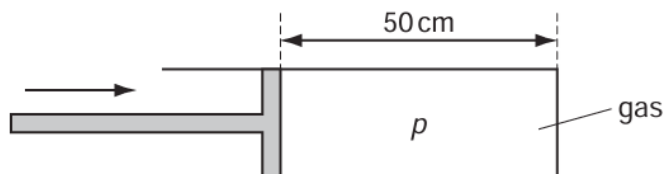


Which statement is correct?

- A** The gas expands.
- B** The molecules get larger.
- C** The piston remains in the same place.
- D** The speed of the molecules decreases.

51. M/J/07/Q12

A gas is trapped inside a cylinder by a movable piston. The length of the gas column is 50 cm and the pressure inside the cylinder is p .



The piston is pushed in a distance of 30 cm, so that the length of the gas column is now 20 cm. The temperature of the gas does not change.

What is the new pressure of the gas?

- A** $0.4p$
- B** $0.6p$
- C** $1.5p$
- D** $2.5p$

52. M/J/07/Q14

What describes the molecular structure of a liquid?

	distance between the molecules	motion of the molecules	strength of forces between the molecules
A	close together	stationary	very strong
B	close together	random	fairly strong
C	far apart	stationary	fairly strong
D	far apart	random	weak

53. O/N/06/Q13

What is a property of both liquids and gases?

- A** They always fill their containers.
- B** They are incompressible.
- C** They can flow.
- D** They have molecules in fixed positions.

54. M/J/06/Q16

A substance consists of particles that are close together and moving past each other at random. The average speed of the particles is gradually increasing.

What best describes the substance?

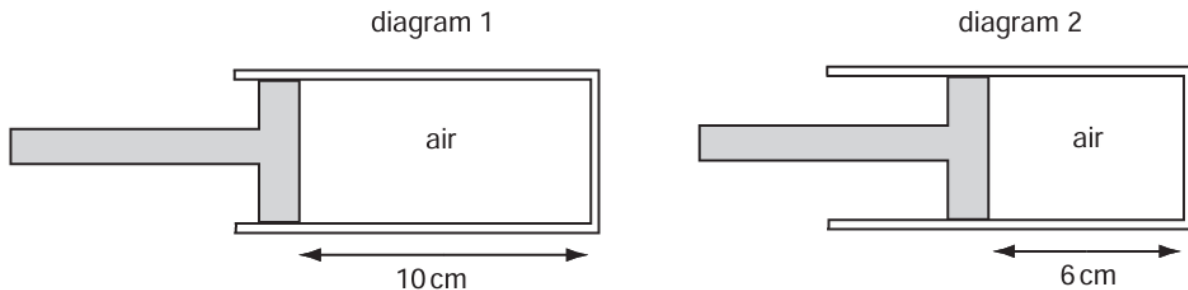
- A** a gas being heated
- B** a liquid being heated
- C** a solid being heated
- D** a solid being melted

55. O/N/04/Q15

Diagram 1 shows some air trapped in a cylinder.

Diagram 2 shows the same air after the piston has been pushed in slowly.

The air in diagram 1 is at atmospheric pressure P_A .



What is the pressure of the air in diagram 2?

- A $\frac{6}{10} \times P_A$
- B P_A
- C $\frac{10}{6} \times P_A$
- D $60 \times P_A$